

Evaluating the Performance of Large Language Models for One Atmosphere Using Automated Extracted Datasets

Shiqin Dai, Qingru Wu,* Haowen Zhang, Yan Wang, and Shuxiao Wang*



Cite This: *Environ. Sci. Technol.* 2026, 60, 725–735



Read Online

ACCESS |



Metrics & More



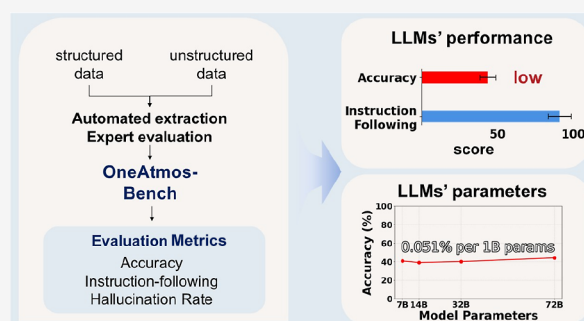
Article Recommendations



Supporting Information

ABSTRACT: Large language models (LLMs) have great potential to improve the efficiency of atmospheric science research and air pollution control decision-making. However, due to the absence of a systematic evaluation benchmark, it remains unclear whether LLMs can be trusted to support core air pollution control tasks such as pollution alarming and mitigation recommendation, which limits their real-world deployment. This study developed a benchmark dataset aligned with the One Atmosphere framework (OneAtmos-Bench), by using a two-stage automated extraction pipeline. Leveraging this dataset, 11 LLMs were evaluated across three key metrics: accuracy, instruction-following (IF), and hallucination rate (HR). Expert validation confirms that the OneAtmos-Bench dataset achieves 95.88% accuracy. The evaluation results indicate that while IF remains consistently strong, both accuracy and hallucination suppression are still constrained, especially for long-tail tasks in the atmospheric domain. Notably, the marginal improvements from scaling up the model size for atmospheric tasks highlight sparse domain adaptation, and common training strategies to enhance reasoning ability, particularly distillation-based supervised fine-tuning, may inadvertently increase the HR. These findings reveal that general-purpose LLMs present reliability challenges in delivering the trustworthy, low-hallucination guidance required for efficient air pollution control decision-making, highlighting the need for environment-specific adaptation.

KEYWORDS: large language models, benchmark, pollution control, OneAtmos-Bench, sparse domain adaptation, post-training



INTRODUCTION

Air pollution represents a critical global challenge, necessitating the integration of artificial intelligence (AI) to advance both atmospheric science research and air pollution management strategies.¹ Among these, air-quality decision-making is a core task that involves a complex chain of processes, such as meteorological analysis, real-time monitoring, atmospheric modeling, mitigation recommendation, and postevent assessment. Many of these tasks are both labor-intensive and time-sensitive, often requiring expert judgment under pressure.² Large language models (LLMs), which are one of the most significant breakthroughs in natural language processing in recent years,^{3,4} are increasingly regarded as promising tools to accelerate the understanding of pollution causes and reduce the response time in emission control.^{5,6} However, the development of domain-specific LLMs, such as those for ecological and environmental applications, has lagged behind the rapid evolution of general-purpose LLMs, as evidenced by the lack of specialized knowledge,⁷ frequent hallucinations, and limited reasoning capabilities in domain-specific tasks.^{8–10} These limitations are particularly concerning in air pollution alarming and emergency management, where minor interpretive errors may lead to delayed interventions and severe public health consequences.^{11,12} Whether LLMs can provide

trustworthy, low-hallucination guidance for real-world environmental decision-making has become a shared concern among government officials and researchers.

A large amount of work has already focused on domain-specific adaptation of LLMs in the atmospheric domain, making important contributions to alleviating hallucinations and enhancing the professionalism and accuracy of LLMs' responses in Earth science applications. ChatClimate¹³ and AirGPT,¹⁴ for example, integrate domain-specific corpora from climate and atmospheric sciences with retrieval-augmented generation (RAG), thereby improving the quality of LLM-based question answering in climate science and air-quality management at a relatively low cost. Similarly, models such as ClimateChat,¹⁵ Jiuzhou,¹⁶ and K2¹⁷ build upon foundation models (e.g., LLaMA-7B) and employ corpora from climate science or Earth sciences, using approaches such as low-rank adaptation (LoRA) or continued pretraining to strengthen

Received: July 28, 2025

Revised: November 16, 2025

Accepted: December 9, 2025

Published: December 31, 2025



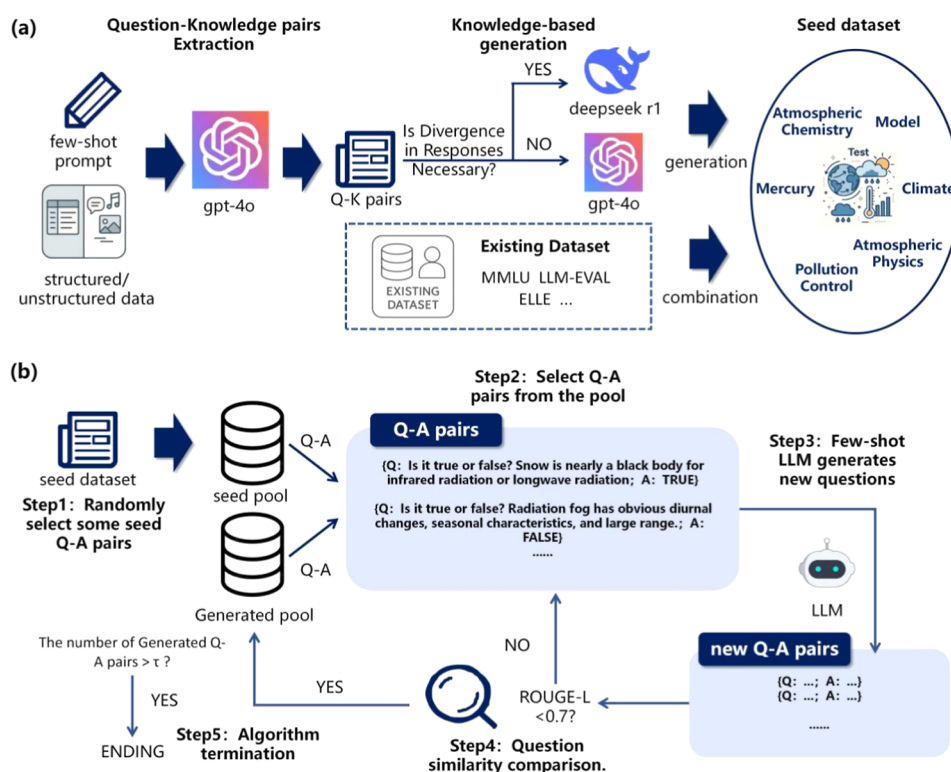


Figure 1. Two-stage automated extraction pipeline for OneAtmos-Bench. (a) Seed dataset generation. A seed dataset covering atmospheric mercury, atmospheric physics, atmospheric chemistry, air-quality model, climate, and pollution control is first constructed. This dataset is then used in (b) iterative self-instruct for automated generation. The algorithm terminates when the number of QA pairs in the generated pool reaches a predefined threshold τ . Q: Question; A: Answering.

domain adaptation. Nevertheless, whether through RAG or other post-training methods, these approaches remain heavily dependent on the base capabilities of LLMs. Therefore, conducting a rigorous evaluation of LLMs' capabilities in atmospheric domains is essential, as it can provide valuable insights for subsequent post-training efforts in this field.

Zhu et al.¹⁸ and Xu et al.¹⁹ conducted evaluations of LLMs in the fields of environmental engineering and water pollution control engineering, respectively. However, current studies have failed to address whether existing LLMs can be effectively applied to atmospheric pollution control. Processes regulating air pollution involve managing emission sources and evaluating their effects, representing a domain that integrates both science and engineering. The coordinated control of multiple air pollutants is recommended to follow the One Atmosphere paradigm,^{20,21} which considers regional air pollution and global convention implementation as two interconnected aspects of a unified atmospheric system, emphasizing their interactions and synergistic effects. This highlights that LLM evaluation under the One Atmosphere framework must be a comprehensive, interdisciplinary effort covering atmospheric mercury, atmospheric chemistry, atmospheric physics, air-quality models, climate, and air pollution control. Meanwhile, existing studies in the atmospheric domain primarily focus on evaluating model responses based on their accuracy by comparing them to reference answers.^{18,22} However, hallucination, referring to the tendency of LLMs to generate inaccurate or fabricated information when answering open-ended questions, remains largely overlooked in current evaluation frameworks, despite its critical impact on the practical applicability of models in the atmospheric domain.^{8,23} A comprehensive, multidimensional

evaluation of LLMs can help uncover their adaptability across various atmospheric tasks, supporting the development of diverse applications such as pollution forecasting, source apportionment, and atmospheric emergency decision-making.

To facilitate the evaluation of LLMs in environmental science and engineering, several recent studies have proposed a series of benchmarks incorporating environmental knowledge, including general knowledge benchmark LLM-EVAL,²⁴ the environmental benchmark ELLE,²⁵ the atmospheric and climate benchmarks AtmosSci-Bench,²² ClimaBench,²⁶ and ClimateX.²⁷ In existing general knowledge or environmental benchmarks, data related to the atmospheric domain typically account for less than one-quarter of the dataset, making them insufficient for comprehensive evaluation. Meanwhile, benchmarks focused on atmospheric and climate sciences often emphasize a single subfield, such as atmospheric dynamics or climate, thus failing to meet the interdisciplinary assessment needs of the One Atmosphere framework. Moreover, the construction of these datasets is typically based on the manual collection of existing atmospheric domain questions^{22,26,28} and expert annotations,²⁷ which is highly labor-intensive. Therefore, the automatic construction of an evaluation dataset targeting air pollution process control, along with the assessment of LLMs through a One Atmosphere-based evaluation framework, is essential for validating their applicability and reliability in supporting the atmospheric domain.

In this study, to evaluate the overall capabilities of LLMs in performing pollution control tasks under the One Atmosphere framework, we propose an automated pipeline for extracting domain-specific QA pairs and construct a benchmark dataset,

(a)					(b) T/F question prompt	
Name	Creator	Size	Launch time	Source types	You are an expert in the field of atmospheric science. Please answer the question: {question} Note: Answer the question in English. Answer directly with "True" or "False".	calculation/generation question prompt You are an expert in the field of atmospheric science. Please answer the question: {question} Note: Answer the question in English. For calculation problems, provide the calculation steps.
qwq-32b	Alibaba Cloud	32B	2025.3	Open source		
qwen2.5-72b-instruct		72B	2024.9	Open source		
qwen2.5-32b-instruct		32B	2024.9	Open source		
qwen2.5-14b-instruct		14B	2024.9	Open source		
qwen2.5-7b-instruct		7B	2024.9	Open source		
qwen-max-latest	DeepSeek	Unknown	2025.1	Closed source		
deepseek-v3		671B	2024.12	Open source		
deepseek-r1-distill-qwen-32b	Zhipu	32B	2025.1	Open source		
glm-4-plus		Unknown	2024.8	Closed source		
gpt-3.5-turbo		~175B	2022.11	Closed source		
gpt-4o	OpenAI	~200B	2024.5	Closed source		

Figure 2. Foundation models and prompt engineering. (a) Foundation models evaluated in this paper. (b) Prompt templates used for answering atmospheric domain questions. {question} is used to sequentially store each question from OneAtmos-Bench, and the entire template is fed into the LLMs during evaluation.

OneAtmos-Bench. Next, we conduct a comprehensive evaluation of LLMs in the atmospheric domain based on a three-dimensional framework encompassing accuracy, hallucination rate (HR), and instruction-following (IF) capability. To further promote the application of LLMs in the atmospheric domain, we investigate the impact of different training strategies and model parameters on the performance of LLMs in this vertical domain. Our work quantitatively assesses the capabilities of general-purpose LLMs in addressing air-quality decision-making tasks under the One Atmosphere framework and provides actionable guidance for selecting and adapting models to deliver trustworthy, low-hallucination outputs in efficient air pollution control decision-making.

METHODS

Automated Construction of OneAtmos-Bench

Evaluating LLMs in vertical domains requires extensive domain-specific QA data. Currently, Self-QA²⁹ and Self-Instruct³⁰ approaches are widely employed for automatically acquiring such data using LLMs. To rapidly and automatically obtain evaluation data in the atmospheric domain, we improved and integrated these two methods and propose a novel pipeline for extracting atmospheric QA data. First, a seed dataset was constructed by combining an existing dataset and knowledge-based generation, utilizing question–knowledge (QK) pairs extracted from literature and reports (Figure 1a). Next, leveraging the seed dataset, we iteratively generated additional QA pairs automatically through in-context learning, exploiting the generative capability of LLMs (Figure 1b). To ensure a structured and hierarchical evaluation, we categorized the questions into three types—true/false questions, calculation questions, and open-ended questions—as well as three difficulty levels: simple, medium, and difficult. Simple questions typically involve factual knowledge, requiring minimal reasoning, whereas difficult questions entail complex computations, reasoning processes, and creative capabilities.

Seed Dataset Generation. We categorized atmospheric domain knowledge into six subdomains: atmospheric mercury, atmospheric chemistry, atmospheric physics, climate, air-quality model, and pollution control. First, we directly

incorporated atmospheric science-related data from general knowledge datasets (e.g., LLM-EVAL²⁴) and environment-specific datasets (e.g., ELLE²⁵) into our seed dataset. Subsequently, structured and unstructured data were collected from relevant literature indexed in the Web of Science as well as authoritative textbooks within the atmospheric domain (Table S1). Considering that directly extracting QA pairs from raw data might result in incomplete information extraction, especially when answering a question requires integrating knowledge across multiple texts, we implemented a strategy of first extracting QK pairs from the collected data, followed by knowledge-based generation utilizing advanced LLMs. GPT-4o, combined with few-shot prompting (Table S2), was employed to extract these QK pairs. Recognizing the varying degrees of divergence required to address different questions, each extracted seed question was annotated by GPT-4o for whether divergent thinking was necessary (details provided in Text S1), guiding the selection of models for knowledge-based generation accordingly. As DeepSeek R1 is recognized for its stronger capacity in divergent thinking,³¹ it was selected for tasks requiring broader reasoning and cross-contextual synthesis. In contrast, GPT-4o was employed for nondivergent tasks due to its widely acknowledged superiority in factual consistency and response faithfulness. The generated QK–answer pairs were reviewed by domain experts before incorporation into the seed dataset.

Iterative Self-Instruct. Based on the seed dataset, new QA pairs were generated through an iterative self-instruct process. First, QA pairs were randomly selected from the seed dataset and stored in a seed pool. In each iteration, a total of eight QA pairs were sampled from the seed pool and the generated pool (only the seed pool was used in the first round, while both pools were used in subsequent rounds). These sampled QA pairs were then provided to LLMs via in-context learning to generate four new QA pairs. To ensure content diversity within the dataset, ROUGE-L³² was used to assess the similarity between each newly generated question and those already existing. ROUGE-L is a metric that measures the longest sequence of words shared in order between two texts, even if they are not consecutive. This allows for a nuanced evaluation of similarity, capturing both syntactic and semantic overlaps

Table 1. Results of Expert Evaluation for OneAtmos-Bench

field	atmospheric mercury	atmospheric physics	atmospheric chemistry	climate	air quality model	pollution control
quantity	115	126	256	126	103	268
simple quantity	28	84	99	105	38	116
medium quantity	38	28	97	14	38	87
difficult quantity	49	14	60	7	27	65
reliability	93.91%	97.62%	94.14%	96.03%	89.32%	100%

between the generated and reference texts. If the similarity score was below the threshold, then the QA pair was added to the generated pool; otherwise, it was returned to the LLM for regeneration. In this study, the threshold was empirically set to 0.7 based on literature reports and repeated experimentation.^{33,34} The iterative process terminated when the number of QA pairs in the generated pool exceeded the predefined threshold τ , resulting in the final atmospheric domain evaluation dataset, OneAtmos-Bench.

Expert Review. To ensure the reliability of the dataset, OneAtmos-Bench was evaluated by experts in the field of atmospheric domain. The evaluation focused on the appropriateness of the difficulty classification and accuracy of the answers. For questions flagged by experts, further verification and necessary corrections were conducted.

Foundation Models and Prompt Engineering

Using OneAtmos-Bench, we evaluated the capabilities of 11 LLMs in the atmospheric domain across three dimensions: accuracy, IF, and HR. The evaluated LLMs include qwq-32b, qwen2.5-72b-instruct, qwen2.5-32b-instruct, qwen2.5-14b-instruct, qwen2.5-7b-instruct,³⁵ qwen-max-latest, deepseek-v3,³⁶ deepseek-r1-distill-qwen-32b,³¹ glm-4-plus,³⁷ gpt-3.5-turbo, and gpt-4o³⁸, as shown in Figure 2a. We selected these models because the Qwen series is a widely adopted choice for fine-tuning and deployment across various vertical domains, while other models such as qwen-max-latest, deepseek-v3, glm-4-plus, and gpt-4o represented the state of the art at the time of publication and are thus of considerable representativeness. Distinct from previous studies, this article primarily focuses on open-source models, given their relatively smaller size and lower fine-tuning costs, which make them more suitable for localized deployment in scenarios with limited computational resources, such as enterprises and government agencies. To investigate the impact of model size and training strategies on domain-specific performance in the atmospheric domain, we selected models with varying parameter scales and training approaches. Moreover, to accommodate the characteristics of different question types, we designed and applied tailored prompts for each, as illustrated in Figure 2b.

Evaluation Metrics and Tools

To comprehensively evaluate the overall performance of the models, we selected three evaluation metrics: Accuracy, IF, and HR. All three metrics were ultimately normalized to a 100-point scale for consistency and comparability. To account for the stochasticity of LLMs, which refers to the variability in performance when the same model responds to the same question multiple times, we conducted three independent runs. The reported results represent the averaged scores across these runs, with error bars plotted to illustrate the variance.

Accuracy. Accuracy measures the proportion of correct predictions or generated responses produced by the model. To account for the varying complexity of different question types, we assigned different weights to each: 1 point for true/false

questions, 3 points for open-ended questions, and 5 points for calculation questions. For true/false and calculation questions, objective evaluation was conducted based on the correctness of the answers. Given the inherent challenges of evaluating open-ended questions, we adopted the recall rate from ROUGE-L³³ as the scoring criterion:

$$\text{recall rate} = \frac{|P \cap R|}{|R|} \quad (1)$$

where P is the number of n -gram of LLM's precision, and R is the number of n -gram of reference answer. Since the recall rate tends to overestimate scores for logically reversed or partially correct answers, we conducted manual verification for factual open-ended questions following the initial automatic evaluation (Text S2). To further investigate the underlying sources of errors in LLM responses, we selected one computational problem and one open-ended problem as case studies, using a strong open-source model (qwq-32b) and a widely used closed-source model (gpt-4o) as examples.

Instruction-Following. IF refers to a model's ability to accurately understand and execute tasks as specified by a user's prompt or command, adhering to the intended meaning, required format, and any constraints provided. For true/false questions, the prompt explicitly included the instruction "answer directly"; therefore, any responses containing redundant or extraneous content were considered deficient in IF. True/false and calculation questions were evaluated manually to assess instruction adherence; if a model's response contained information beyond "True" or "False", the question was marked as incorrect. Open-ended questions were automatically evaluated using GPT-4o through prompt engineering, as detailed in Table S2.

Hallucination Rate. HR is typically defined as the proportion of hallucinated claims relative to the total number of claims in a given response. In this study, we employ RefChecker³⁹ as the hallucination evaluation tool. RefChecker operates by decomposing the LLM-generated responses into knowledge triples (subject, predicate, and object) and evaluating the factuality of each triple against a provided reference text, thereby enabling the precise detection of hallucinations.

RESULTS AND DISCUSSION

OneAtmos-Bench

We developed OneAtmos-Bench, a benchmark tailored to the One Atmosphere framework, by constructing a two-stage automated extraction pipeline (Figure 1) to derive QA pairs from atmospheric domain textbooks and scholarly literature (Table S1), each of which was subsequently reviewed and validated by domain experts. The OneAtmos-Bench dataset comprises a total of 994 validated QA pairs, including 166 true/false questions, 33 calculation questions, and 795 open-ended questions. Compared with AtmosSci-Bench, which

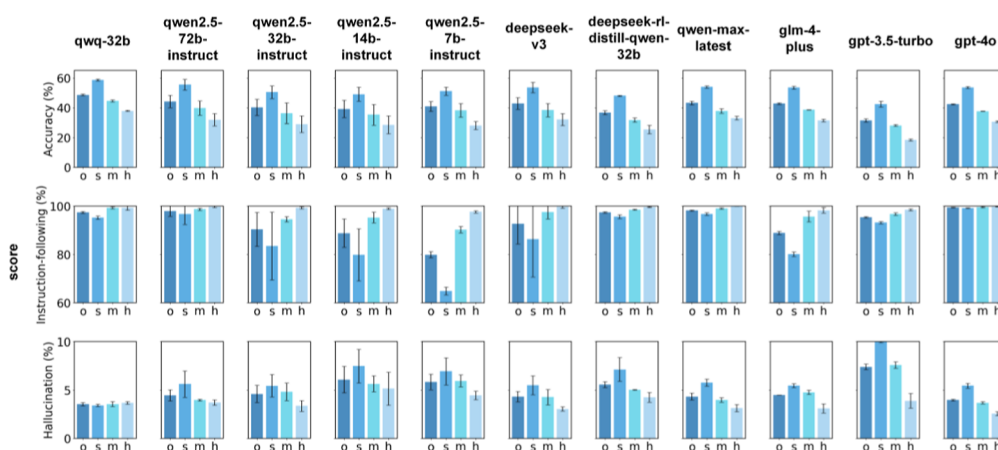


Figure 3. Performance of selected LLMs in terms of accuracy, IF, and hallucination. Scores for each metric are shown at three difficulty levels: s (simple), m (medium), and h (hard), as well as the overall performance (o).

includes only multiple-choice questions, it places greater emphasis on the evaluation of open-ended questions and involves a higher level of difficulty.

Table 1 presents the number of QA pairs across different subdomains and the corresponding expert-verified accuracy rates. In comparison with previous benchmarks in the atmospheric domain, OneAtmos-Bench is oriented toward the One Atmosphere concept and introduces evaluations across multiple dimensions in the atmospheric domain, including pollutant-specific analysis (e.g., atmospheric mercury), scientific understanding (e.g., atmospheric chemistry), and control strategies (e.g., multipollutant control). The overall accuracy of the automated data extraction method reached 95.88%, as verified through expert review, indicating its reliability and potential applicability. Experts also provided relevant references for problematic answers based on their domain expertise. Incorporating this feedback, we corrected the instances of hallucination and subjected the dataset to an additional round of expert review, thereby ensuring its overall reliability. However, the accuracy in certain subdomains, such as the atmospheric mercury and air-quality model, was comparatively lower. This may be attributed to outdated literature used during seed dataset generation, as well as hallucinations introduced by the model during the iterative self-instruct process.⁴⁰ The subjectivity of expert evaluation is also a noteworthy issue. Although we invited domain experts from various subdomains of atmospheric science to review the data, ensuring a certain level of reliability, expert assessments of open-ended questions inevitably carry personal bias. Enhancing the timeliness and reliability of both structured and unstructured data sources, along with mitigating hallucination of LLMs during self-instructing, is the key to improving the robustness of this pipeline in future works.

Evaluation of LLMs' Performance in the Atmospheric Domain

Figure 3 shows different LLMs' performance in the atmospheric domain. In general, while closed-source models in the atmospheric domain generally possess larger parameter scales and are often assumed to deliver superior performance, our evaluation reveals that their advantage is limited to IF, with no significant gains observed in either accuracy or hallucination suppression. For example, among all closed-source models, gpt-4o (accuracy = 42.38%, HR = 3.98%) and qwen-max-latest (accuracy = 43.18%, HR = 4.34%) demonstrated relatively

strong performance. However, their accuracy and HR in the atmospheric domain remained lower than those of the best-performing open-source model in our evaluation, qwq-32b (accuracy = 48.58%, HR = 3.56%).

Notably, none of the evaluated models surpassed 60% accuracy or achieved an HR below 3%. These results further reveal that current general-purpose LLMs suffer from both knowledge gaps in domain-specific concepts and a lack of rigor in logical reasoning (Tables S6 and S7). In contrast, LLMs performed relatively well in IF, especially models in the GPT family, with GPT-3.5-Turbo scoring 95.45% and GPT-4o reaching as high as 99.42%, indicating near-perfect adherence to task instructions. The heterogeneity across capability dimensions, coupled with low scores on domain-specific metrics, highlights the current limitations of LLMs in addressing atmospheric science tasks and underscores the need for a deeper investigation into both task difficulty adaptation and subdomain-specific generalization. At the same time, we observed that although the differences across repeated experiments were generally small, demonstrating the robustness of our methodology, certain models (e.g., qwen-2.5 and deepseek-v3) exhibited substantial variability across experiments conducted at different times. This can be attributed to the fact that even models with the same name may undergo updates over time (see Alibaba Cloud official logs: <https://www.alibabacloud.com/help/zh/model-studio/model-depreciation>, <https://help.aliyun.com/zh/model-studio/model-announcements>). With these parameter updates, the performance of LLMs in the atmospheric domain showed a notable decline (Figure S1). This suggests that parameter updates or training strategies targeting general-domain capabilities may not adequately address the sparse adaptation problem in vertical domains such as the atmospheric domain.

Exploring the adaptability of LLMs to domain-specific tasks of varying difficulty serves as an important means of characterizing their ability to address challenges in vertical scientific domains. The results indicate that as the question difficulty increased, model accuracy generally declined, while IF performance improved, and HR decreased. This inverse trend may be attributed to the emergence of chain-of-thought (CoT), which encourages step-by-step reasoning in complex tasks. By decomposing intricate reasoning or calculation questions, CoT techniques may enhance instruction comprehension and promote more factually grounded responses but

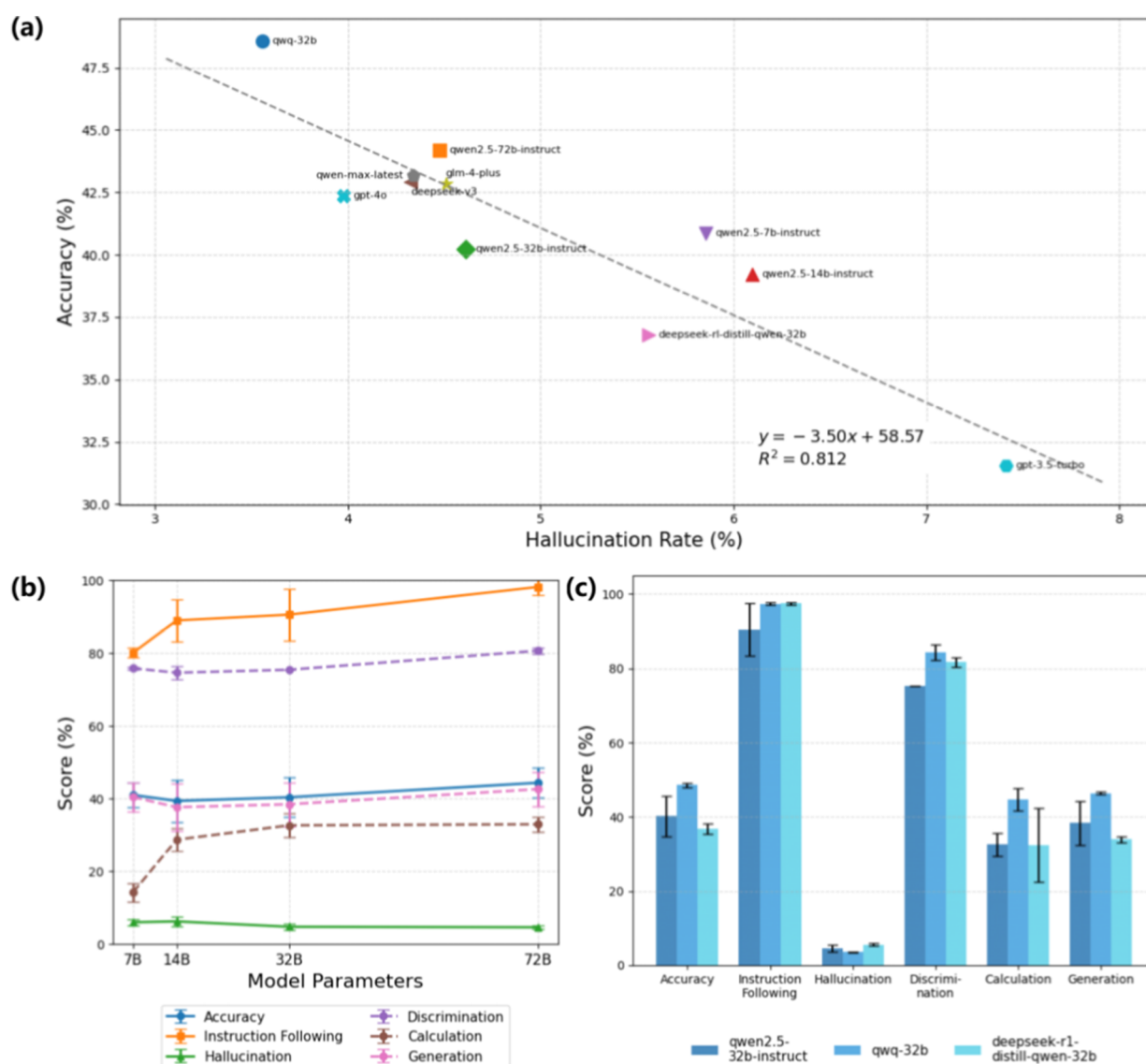


Figure 4. Relationships among evaluation metrics and the effects of model size and training strategy on the performance of LLMs. (a) Relationship between accuracy and HR. The dashed lines represent the trends of HR and accuracy. (b) Sensitivity of different evaluation metrics to model parameter scaling. The solid lines represent three evaluation metrics in this paper (accuracy, IF, and HR), while the dashed lines indicate accuracy results categorized by question types (discrimination, calculation, and generation, expressed as percentages). (c) Impact of different training strategies on model performance in the atmospheric domain. Discrimination, calculation, and generation represent the accuracy scores of different LLMs when answering true/false questions, calculation questions, and open-ended questions, respectively.

may simultaneously lead to greater deviations from reference answers. We hypothesize that LLMs may employ different reasoning mechanisms when confronted with tasks of varying difficulty and that such shifts in reasoning strategy can influence their performance in the atmospheric domain.

Considering the potential issue of uneven corpus coverage across different subdomains, we analyzed model performance across multiple subtopics, treating accuracy as a primary metric reflecting the closeness of LLM responses to expert-validated answers in each subdomain (Figure S2). In general, LLMs perform best on “climate” (scores: 44.57–59.49%, average: 53.12%) and exhibit the poorest performance in “atmospheric mercury” (scores: 23.49–40.96%, average: 32.90%). LLMs tend to provide more accurate and factual answers in the “climate” domain, whereas in the “atmospheric mercury” domain, their performance is often weaker. Since the difficulty distribution of questions varies across subdomains (Table 1),

accuracy scores alone make it challenging to disentangle whether weaker performance stems from gaps in training corpora or from the inherent complexity of the questions. Notably, in subdomains with lower scores, particularly *atmospheric mercury*, we observed frequent misuse of technical terminology. For instance, most models failed to correctly identify the abbreviation GOM (gaseous oxidized mercury), resulting in inaccurate responses. This indicates that in certain subdomains with more complex systems, the limited availability of high-quality training data has led to uneven distribution of commercial LLM training corpora across scientific subfields.^{41–43} These findings highlight the need to supplement domain-specific data on under-represented areas, such as “atmospheric mercury”, when fine-tuning LLMs for atmospheric applications.

Overall, differences in reasoning mechanisms and corpus structure may contribute to the heterogeneity observed across

different capability dimensions of LLMs. Therefore, understanding the relationships among evaluation metrics can help identify effective strategies for improving the multifaceted capabilities of LLMs. As shown in Figure 4a, a strong negative linear relationship ($R^2 = 0.812$) was observed, indicating that models with higher accuracy are generally less prone to hallucination. This finding aligns with prior work (e.g., Tang et al.⁴⁴) and suggests that robust internal knowledge representation and sound reasoning mechanisms can suppress unsupported content generation. This performance ceiling highlights post-training strategies such as supervised finetuning (SFT) and RAG may not only improve accuracy⁴⁵ but also contribute to reducing hallucination by encouraging source-grounded responses, particularly critical in high-stake domains such as atmospheric pollution control.

Impact of Model Iteration on Model Performance

When iterating general-purpose LLMs, developers often adopt larger model parameters and alternative training strategies. To examine whether such iterations substantially enhance performance in vertical domains like the atmospheric domain, we investigated the impact of two key factors on LLM performance: model parameters and training strategy.

Model Parameters. We first evaluated the scaling law that characterizes how the model capability scales with parameter size within the atmospheric domain. Overall, as the number of model parameters increases, both accuracy and IF improve, while HR decreases. This trend is nonlinear, especially for accuracy and HR (Figure 4b). The improvement in IF is more pronounced than that in accuracy, suggesting that the growth of natural language understanding capabilities outpaces the accumulation of atmospheric domain knowledge.

Surprisingly, while the model size tends to enhance discriminative and, especially, mathematical reasoning abilities in the atmospheric domain, such improvements are not always consistent. The performance on generation-type tasks in the atmospheric domain exhibited almost no improvement with an increasing model size, and even the minor gains observed in smaller models fell within the natural variance of LLMs. Meanwhile, the decrease in HR with an increasing model size is also not consistent. The increase in knowledge capacity does not show a strictly positive correlation with response factuality, which appears to contradict the scaling law. We hypothesize that this is related to the nature of atmospheric tasks as long-tail tasks, characterized by small sample sizes, limited training resources, high task complexity, and broad coverage, which occur infrequently in training data but are common in real-world applications. The atmospheric domain is marked by a wide semantic scope and complex reasoning chains, and its long-tail nature leads to the dilution of atmospheric knowledge during the pretraining of commercial LLMs, resulting in limited domain-specific generalization. This highlights the need to construct more fine-grained, CoT-based, and cross-modal QA pairs for domain-specific SFT, in order to improve the model's fitting ability in the atmospheric domain.

Training Strategy. At present, two mainstream approaches are applied to improve the reasoning capabilities of foundation LLMs: reinforcement learning (RL)-based methods (e.g., qwq-32b) and distillation-based SFT using CoT data generated by the teacher model (e.g., deepseek-rl-distill-qwen-32b). To investigate the impact of training strategies on model performance in the atmospheric domain, we compared models

with identical parameter sizes (32B) but different reasoning enhancement methods.

Figure 4c shows that reasoning-augmented models exhibit notable improvements over foundation models (qwen2.5-32b-instruct) in discriminative tasks, numerical reasoning, and IF performance. However, while RL-based models demonstrated improved performance in generative tasks (with accuracy increasing from 40.23% to 48.58%), distillation-based models exhibited a marked decline (with accuracy decreasing from 40.23% to 36.79%). Catastrophic interference,^{18,46} where LLMs may forget previously acquired world knowledge when learning new information, occurs with SFT using distilled CoT data, while RL-based training appears to encourage the emergence of CoT reasoning to preserve factual consistency in generative outputs.^{47,48}

However, reasoning-augmented models, particularly those trained with distillation-based SFT, have the potential to increase HR. Overthinking, wherein excessive reasoning increases the likelihood of generating unsupported content, might contribute. Notably, distillation-based models exhibit a greater tendency to increase HR compared to RL-based models, which may result from a stronger emphasis on stylistic coherence during SFT at the expense of factual consistency. Moreover, the teacher model of deepseek-rl-distill-qwen-32b, DeepSeek R1, inherently exhibits a relatively high HR.⁴⁹ Consequently, during distillation-based SFT, the student model may inherit and even amplify the hallucination tendencies of the teacher model.

Overall, this study finds that although increasing model parameters and adopting new training strategies can enhance the capabilities of general-purpose LLMs for standard tasks, such approaches alone fail to achieve a qualitative leap in performance for long-tail tasks in the atmospheric domain—and even in the broader environmental field owing to its long-tail nature. Therefore, targeted post-training remains necessary, as domain-adapted models for long-tail tasks are unlikely to be replaced by the continual iterations of general-purpose LLMs.

Strengths and Limitations

Although the automated data extraction pipeline proposed in this study substantially reduces the manual effort required to construct domain-specific datasets, the processes of dataset validation and expert review still demand considerable time and human resources. As the seed dataset is primarily derived from manually selected structured and unstructured sources, the origin and degree of systematization of the data vary across different subdomains, inevitably leading to biases and corpus limitations in the construction of evaluation datasets. For example, in the case of “atmospheric mercury”, due to the limited availability of relevant textbooks, the seed dataset was mainly composed of literature-derived content, whereas for “atmospheric chemistry” and “atmospheric physics”, which have more mature knowledge systems, textbook-based extraction was more prevalent. The inherent differences in credibility between the literature and textbooks introduce additional variability. Beyond these data-related factors, cross-domain comparability also poses challenges: a “hard” question in atmospheric mercury may not be directly comparable to a “hard” question in climate, where knowledge is more systematically consolidated. Distinguishing between question complexity and the availability of pretraining corpora within subdomains remains a pressing challenge.

Meanwhile, the automatic evaluation approach adopted in this study does not fully address the ambiguity of the evaluation standards and behavioral uncertainty of models in open-ended question assessments. Metrics such as accuracy, which rely on textual overlap, are insufficient for evaluating the semantic appropriateness of answers to open-ended questions. Despite the inclusion of human correction, semantic deviation remains an issue for divergent questions. Furthermore, using recall as a metric for accuracy may disadvantage models that produce concise and focused responses.

Additionally, beyond post-training approaches, RAG is often regarded as providing a more direct and controllable means of injecting knowledge and mitigating hallucinations. The performance of RAG is typically influenced by the retrieval method (e.g., sparse versus dense retrieval) as well as the quality of the knowledge base, underscoring the need for systematic evaluations to clarify how different RAG construction strategies affect LLM performance in the atmospheric domain. This line of investigation was not addressed in the present study, and systematic efforts in this direction remain necessary for future research.

In terms of evaluation metrics, this study employed accuracy, IF, and HR as indicators for assessing LLM performance in the atmospheric domain. However, these metrics may be somewhat limited in fully capturing performance in atmospheric applications. For example, certain questions require cross-domain knowledge synthesis and hypothesis generation. If LLMs are treated merely as fact retrievers and precise calculators, then their responses to such questions are often inadequate. At present, there is a lack of quantitative metrics to evaluate the model's capacity for "divergent thinking". In the future, it will be important to establish a more comprehensive evaluation framework capable of capturing differences in LLMs' reasoning styles and answer-generation approaches.

Nevertheless, our framework can be readily extended to evaluate and compare future LLMs in the atmospheric domain, and we encourage its use for systematically assessing the capabilities of LLMs. Future research on LLMs should place greater emphasis on application-oriented scenarios. For instance, the impact of different domain or general data mixing ratios during fine-tuning can enable LLMs to serve the needs of atmospheric tasks like pollution alarming, source appointment, and proposed measures more effectively.

ENVIRONMENTAL IMPLICATIONS

Fundamental Data Infrastructure for Atmospheric Domain

As highlighted in this study, targeted post-training is necessary for long-tail tasks in the atmospheric and environmental domains, yet data scarcity continues to hinder the advancement and deployment of LLMs in these vertical areas. High-quality QA datasets, encompassing various aspects of atmospheric science, such as emissions, atmospheric chemistry, atmospheric physics, and health and climate benefits, enable sufficient domain-specific post-training. This facilitates the integration of scientific knowledge and enhances reasoning capabilities within the atmospheric domain, thereby partially mitigating the challenges of sparse domain adaptation. In this study, a semiautomated data extraction pipeline was developed to construct QA datasets from unstructured sources such as reports and literature in the atmospheric domain. This approach replaces the labor-intensive process of manual expert annotation in LLM training data construction, significantly

reducing human effort and providing a practical reference for dataset development in long-tail tasks. However, due to inherent hallucinations in LLMs, the temporal limitations of source materials, and the challenges of cross-paragraph and cross-document information extraction, the resulting QA datasets may inevitably contain inaccuracies or outdated content. Thus, expert review remains necessary.

Insights for Building Multiagent Systems of Atmospheric Domain

LLM-based autonomous agents developed rapidly and received widespread and high attention.⁵⁰ Multiagent systems based on the collaboration of multiple LLM-based autonomous agents can achieve intelligent execution of domain-specific workflows in atmospheric domain, such as pollution case retrieval and pollution forecasting, through mechanisms including memory, tool learning methods, and inter-agent communication.^{51,52} This study provides important insights into the selection of LLM-based agents for different atmospheric tasks based on evaluations across three key metrics. For tasks with high IF requirements, such as the automatic extraction of formulas, model configurations, conclusions, and uncertainty information from atmospheric reports, closed-source models such as GPT-4o and qwen-max-latest are suitable choices. For tasks where hallucinations can have critical consequences, such as pollution process analysis and forecasting, where even minor factual inaccuracies may lead to erroneous interpretations of pollution dynamics, foundation models with larger parameter sizes, such as gpt-4o, deepseek-v3 application programming interface, or open-source models above 32B, are recommended. Reasoning models should be avoided in such cases due to the observed increase in both the HR and response latency. In this sense, such evaluations could guide the design of more effective model team compositions for atmospheric tasks, shifting the focus from solely selecting the highest scoring generalist model toward understanding and composing models, which are more aligned with the emerging paradigm of AI agent collaboration.

In the future, building multiagent systems based on these suggestions, and integrating domain-specific tools like deepRSM,⁵³ presents a promising direction for automating atmospheric process management, including air-quality forecast, source apportionment, and clean air measure screening.

Insights for LLMs' Post-Training in the Atmospheric Domain

Fine-tuning LLMs using domain-specific datasets is a common approach to enhance their adaptability in vertical domains. When adapting LLMs to domain-specific tasks, it is essential to incorporate knowledge from various subdomains of the atmospheric domain and conduct targeted training for weaker areas (e.g., atmospheric mercury). To mitigate catastrophic interference introduced by SFT, specialized datasets in the atmospheric domain should be combined with commonsense reasoning datasets to prevent excessive degradation of general-domain capabilities. Following fine-tuning, model performance can be evaluated on the open-source dataset OneAtmos-Bench presented in this study, thereby assessing the extent of capability improvements.

In this study, we investigated the effects of model parameters and training strategies on the performance of LLMs in the atmospheric domain, providing important references for fine-tuning LLMs in this field. From the perspective of model parameters, larger models generally possess greater knowledge

capacity and stronger reasoning abilities but also require significantly more computational resources. We argue that selecting a 32B-scale model represents a balanced and cost-effective choice, as it can be deployed on a single A100-80GB GPU (requiring approximately 61.6 GB BF16 memory for a 1-token input and generating ~ 17.5 tokens/s),⁵⁴ and has already become a common option for deployment in vertical domains.^{55,56} Moreover, its reasoning and computational performance in the atmospheric domain are comparable to those of a 72B-scale model, with only minor differences in generative capability, while the fine-tuning cost of the 72B model is substantially higher (approximately four times greater in practice). Regarding training strategies, although distilling strong reasoning models is an effective method to enhance reasoning capabilities in the atmospheric domain, it may lead to catastrophic interference with domain knowledge and an increased HR. To effectively train LLMs for reasoning in the atmospheric domain, a pipeline that combines SFT and RL is recommended. RL facilitates the emergence of CoT reasoning, enhancing the model's reasoning capability, while SFT using atmospheric datasets mixed with general knowledge helps shape the model's stylistic preferences and improves its domain adaptability.

■ ASSOCIATED CONTENT

Data Availability Statement

Benchmark datasets OneAtmos-Bench (including question, subdomain, difficulty, divergence, and reference answer) is publicly available on GitHub (<https://github.com/BrandNewJimZhang/OneAtmosphere-LLM>). In addition, the LLMs' responses evaluated in this paper have been included in the same repository. The dataset will be continuously updated.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.5c10297>.

Sources of data, technical details of obtaining the OneAtmos-Bench dataset and question examples, specific details of evaluating different LLMs, comparison of results from repeated evaluations, specific results of the three evaluation metrics for LLMs, and a case study of CoT for LLMs (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Qingru Wu — School of Environment, State Key Laboratory of Regional Environment and Sustainability, School of Environment, Tsinghua University, Beijing 100084, P. R. China; State Environmental Protection Key Laboratory of Sources and Control of Air Pollution Complex, Beijing 100084, P. R. China; orcid.org/0000-0003-3381-4767; Email: qrwu@tsinghua.edu.cn

Shuxiao Wang — School of Environment, State Key Laboratory of Regional Environment and Sustainability, School of Environment, Tsinghua University, Beijing 100084, P. R. China; State Environmental Protection Key Laboratory of Sources and Control of Air Pollution Complex, Beijing 100084, P. R. China; orcid.org/0000-0001-9727-1963; Email: shxwang@tsinghua.edu.cn

Authors

Shiqin Dai — School of Environment, State Key Laboratory of Regional Environment and Sustainability, School of Environment, Tsinghua University, Beijing 100084, P. R. China; State Environmental Protection Key Laboratory of Sources and Control of Air Pollution Complex, Beijing 100084, P. R. China

Haowen Zhang — School of Environment, State Key Laboratory of Regional Environment and Sustainability, School of Environment, Tsinghua University, Beijing 100084, P. R. China; State Environmental Protection Key Laboratory of Sources and Control of Air Pollution Complex, Beijing 100084, P. R. China

Yan Wang — College of Electromechanical Engineering, Qingdao University of Science and Technology, Qingdao 266061, China

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.est.5c10297>

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This study was jointly funded by National Natural Science Foundation of China (22222604), National Key Research and Development Program (2023YFC3709403), special fund of State Key Laboratory of Regional Environment and Sustainability (24Y02ESPCT), and Tsinghua-Toyota Joint Research Institute Interdisciplinary Program. We thank the following teams, listed in alphabetical order by the first names of their team leaders, for their contributions to the development of the OneAtmos-Bench benchmark dataset: Da Zhang (Tsinghua University), Jiandong Wang (Nanjing University of Information Science and Technology), Lu Shen (Peking University), Lei Zhang (Nanjing University), Song Guo (Peking University), Yicong He (Tsinghua University), and Yuxi Liu (Beijing University of Technology).

■ REFERENCES

- (1) Silva, S. J.; Evans, M. Artificial Intelligence and Machine Learning in Atmospheric Chemistry. *ACS EST Air* **2024**, *1* (5), 330–331.
- (2) Henneman, L. R. F.; Liu, C.; Mulholland, J. A.; Russell, A. G. Evaluating the Effectiveness of Air Quality Regulations: A Review of Accountability Studies and Frameworks. *J. Air Waste Manage. Assoc.* **2017**, *67*, 144–172.
- (3) Zhao, W. X.; Zhou, K.; Li, J.; Tang, T.; Wang, X.; Hou, Y.; Min, Y.; Zhang, B.; Zhang, J.; Dong, Z.; Du, Y.; Yang, C.; Chen, Y.; Chen, Z.; Jiang, J.; Ren, R.; Li, Y.; Tang, X.; Liu, Z.; Liu, P.; Nie, J.-Y.; Wen, J.-R. A Survey of Large Language Models. *arXiv* **2023**, arXiv:2303.18223.
- (4) Minaee, S.; Mikolov, T.; Nikzad, N.; Chenaghlu, M.; Socher, R.; Amatriain, X.; Gao, J. Large Language Models: A Survey. *arXiv* **2024**, arXiv:2402.06196.
- (5) Rillig, M. C.; Ågerstrand, M.; Bi, M.; Gould, K. A.; Sauerland, U. Risks and Benefits of Large Language Models for the Environment. *Environ. Sci. Technol.* **2023**, *57* (9), 3464–3466.
- (6) Wu, Y.; Xu, M.; Liu, S. Generative Artificial Intelligence: A New Engine for Advancing Environmental Science and Engineering. *Environ. Sci. Technol.* **2024**, *58* (40), 17524–17528.
- (7) Ling, C.; Zhao, X.; Lu, J.; Deng, C.; Zheng, C.; Wang, J.; Chowdhury, T.; Li, Y.; Cui, H.; Zhang, X.; Zhao, T.; Panalkar, A.; Mehta, D.; Pasquali, S.; Cheng, W.; Wang, H.; Liu, Y.; Chen, Z.; Chen, H.; White, C.; Gu, Q.; Pei, J.; Yang, C.; Zhao, L. Domain

Specialization as the Key to Make Large Language Models Disruptive: A Comprehensive Survey. *arXiv* **2024**, arXiv:2305.18703.

(8) Huang, L.; Yu, W.; Ma, W.; Zhong, W.; Feng, Z.; Wang, H.; Chen, Q.; Peng, W.; Feng, X.; Qin, B.; Liu, T. A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions. *ACM Trans. Inf. Syst.* **2025**, *43* (2), 1–55.

(9) Zhang, F.; Yu, P.; Yi, B.; Zhang, B.; Li, T.; Liu, Z. Prompt-Guided Internal States for Hallucination Detection of Large Language Models. *arXiv* **2025**, arXiv:2411.04847.

(10) Wang, Z.; Yu, J.; Ma, D.; Chen, Z.; Wang, Y.; Li, Z.; Xiong, F.; Wang, Y.; E, W.; Tang, L.; Zhang, W. RARE: Retrieval-Augmented Reasoning Modeling. *arXiv* **2025**, arXiv:2503.23513.

(11) Kelly, F. J.; Fuller, G. W.; Walton, H. A.; Fussell, J. C. Monitoring Air Pollution: Use of Early Warning Systems for Public Health. *Respirology* **2012**, *17* (1), 7–19.

(12) Li, R.; Zhen, Z.; Li, X.; Miao, H.; Wei, X. An Air Pollution Early-Warning System with the Mechanism of Dynamic Evaluation. *Urban Clim.* **2025**, *62*, 102531.

(13) Vaghefi, S. A.; Stambach, D.; Muccione, V.; Bingler, J.; Ni, J.; Kraus, M.; Allen, S.; Colesanti-Senni, C.; Wekhof, T.; Schimanski, T.; Gostlow, G.; Yu, T.; Wang, Q.; Webersinke, N.; Huggel, C.; Leipold, M. ChatClimate: Grounding Conversational AI in Climate Science. *Commun. Earth Environ.* **2023**, *4* (1), 480.

(14) Song, J.; Ma, C.; Ran, M. AirGPT: Pioneering the Convergence of Conversational AI with Atmospheric Science. *npj Clim. Atmos. Sci.* **2025**, *8* (1), 179.

(15) Chen, Z.; Wang, X.; Liao, Y.; Lin, M.; Bai, Y. ClimateChat: Designing Data and Methods for Instruction Tuning LLMs to Answer Climate Change Queries. *arXiv* **2025**, arXiv:2506.13796.

(16) Chen, Z.; Lin, M.; Zang, M.; Wang, Z.; Li, J.; Bai, Y. Open Foundation Language Models and Effective Pre-Training Framework for Geoscience. *Int. J. Digital Earth* **2025**, *18* (1), 2449708.

(17) Deng, C.; Zhang, T.; He, Z.; Xu, Y.; Chen, Q.; Shi, Y.; Fu, L.; Zhang, W.; Wang, X.; Zhou, C.; Lin, Z.; He, J. K2: A Foundation Language Model for Geoscience Knowledge Understanding and Utilization. *arXiv* **2023**, arXiv:2306.05064.

(18) Zhu, J.-J.; Yang, M.; Jiang, J.; Bai, Y.; Chen, D.; Ren, Z. J. Enabling GPTs for Expert-Level Environmental Engineering Question Answering. *Environ. Sci. Technol. Lett.* **2024**, *11* (12), 1327–1333.

(19) Xu, B.; Li, Z.; Yang, Y.; Wu, G.; Wang, C.; Tang, X.; Li, Y.; Wu, Z.; Su, Q.; Shi, X.; Yang, Y.; Tong, R.; Wen, L.; Ng, H. Y. Evaluating and Advancing Large Language Models for Water Knowledge Tasks in Engineering and Research. *Environ. Sci. Technol. Lett.* **2025**, *12* (3), 289–296.

(20) Akimoto, H. Global Air Quality and Pollution. *Science* **2003**, *302* (5651), 1716–1719.

(21) Zhao, B.; Wang, S.; Hao, J. Challenges and Perspectives of Air Pollution Control in China. *Front. Environ. Sci. Eng.* **2024**, *18* (6), 68.

(22) Li, C.; Deng, W.; Lu, M.; Yuan, B. AtmosSci-Bench: Evaluating the Recent Advance of Large Language Model for Atmospheric Science. *arXiv* **2025**, arXiv:2502.01159.

(23) Rawte, V.; Sheth, A.; Das, A. A Survey of Hallucination in Large Foundation Models. *arXiv* **2023**, arXiv:2309.05922.

(24) Lin, Y.-T.; Chen, Y.-N. LLM-Eval: Unified Multi-Dimensional Automatic Evaluation for Open-Domain Conversations with Large Language Models. *arXiv* **2023**, arXiv:2305.13711.

(25) Guo, J.; Li, N.; Xu, M. Environmental Large Language Model Evaluation (ELLE) Dataset: A Benchmark for Evaluating Generative AI Applications in Eco-Environment Domain. *arXiv* **2025**, arXiv:2501.06277.

(26) Thulke, D.; Gao, Y.; Pelser, P.; Brune, R.; Jalota, R.; Fok, F.; Ramos, M.; van Wyk, L.; Nasir, A.; Goldstein, H.; Tragemann, T.; Nguyen, K.; Fowler, A.; Stanco, A.; Gabriel, J.; Taylor, J.; Moro, D.; Tsybalov, E.; de Waal, J.; Matusov, E.; Yaghi, M.; Shihadah, M.; Ney, H.; Dugast, C.; Dotan, J.; Erasmus, D. ClimateGPT: Towards AI Synthesizing Interdisciplinary Research on Climate Change. *arXiv* **2024**, arXiv:2401.09646.

(27) Lacombe, R.; Wu, K.; Dilworth, E. ClimateX: Do LLMs Accurately Assess Human Expert Confidence in Climate Statements? *arXiv* **2023**, arXiv:2311.17107.

(28) Bulian, J.; Schäfer, M. S.; Amini, A.; Lam, H.; Ciaramita, M.; Gaiarin, B.; Hübscher, M. C.; Buck, C.; Mede, N. G.; Leipold, M.; Strauß, N. Assessing Large Language Models on Climate Information. *arXiv* **2024**, arXiv:2310.02932.

(29) Zhang, X.; Yang, Q. Self-QA: Unsupervised Knowledge Guided Language Model Alignment. *arXiv* **2023**, arXiv:2305.11952.

(30) Wang, Y.; Kordi, Y.; Mishra, S.; Liu, A.; Smith, N. A.; Khoshabi, D.; Hajishirzi, H. Self-Instruct: Aligning Language Models with Self-Generated Instructions. *arXiv* **2023**, arXiv:2212.10560.

(31) Guo, D.; Yang, D.; Zhang, H.; Song, J.; Zhang, R.; Xu, R.; Zhu, Q.; Ma, S.; Wang, P.; Bi, X.; Zhang, X.; Yu, X.; Wu, Y.; Wu, Z. F.; Gou, Z.; Shao, Z.; Li, Z.; Gao, Z.; Liu, A.; Xue, B.; Wang, B.; Wu, B.; Feng, B.; Lu, C.; Zhao, C.; Deng, C.; Zhang, C.; Ruan, C.; Dai, D.; Chen, D.; Ji, D.; Li, E.; Lin, F.; Dai, F.; Luo, F.; Hao, G.; Chen, G.; Li, G.; Zhang, H.; Bao, H.; Xu, H.; Wang, H.; Ding, H.; Xin, H.; Gao, H.; Qu, H.; Li, H.; Guo, J.; Li, J.; Wang, J.; Chen, J.; Yuan, J.; Qiu, J.; Li, J.; Cai, J. L.; Ni, J.; Liang, J.; Chen, J.; Dong, K.; Hu, K.; Gao, K.; Guan, K.; Huang, K.; Yu, K.; Wang, L.; Zhang, L.; Zhao, L.; Wang, L.; Zhang, L.; Xu, L.; Xia, L.; Zhang, M.; Zhang, M.; Tang, M.; Li, M.; Wang, M.; Li, M.; Tian, N.; Huang, P.; Zhang, P.; Wang, Q.; Chen, Q.; Du, Q.; Ge, R.; Zhang, R.; Pan, R.; Wang, R.; Chen, R. J.; Jin, R. L.; Chen, R.; Lu, S.; Zhou, S.; Chen, S.; Ye, S.; Wang, S.; Yu, S.; Zhou, S.; Pan, S.; Li, S. S.; Zhou, S.; Wu, S.; Ye, S.; Yun, T.; Pei, T.; Sun, T.; Wang, T.; Zeng, W.; Zhao, W.; Liu, W.; Liang, W.; Gao, W.; Yu, W.; Zhang, W.; Xiao, W. L.; An, W.; Liu, X.; Wang, X.; Chen, X.; Nie, X.; Cheng, X.; Liu, X.; Xie, X.; Liu, X.; Yang, X.; Li, X.; Su, X.; Lin, X.; Li, X. Q.; Jin, X.; Shen, X.; Chen, X.; Sun, X.; Wang, X.; Song, X.; Zhou, X.; Wang, X.; Shan, X.; Li, Y. K.; Wang, Y. Q.; Wei, Y. X.; Zhang, Y.; Xu, Y.; Li, Y.; Zhao, Y.; Sun, Y.; Wang, Y.; Yu, Y.; Zhang, Y.; Shi, Y.; Xiong, Y.; He, Y.; Piao, Y.; Wang, Y.; Tan, Y.; Ma, Y.; Liu, Y.; Guo, Y.; Ou, Y.; Wang, Y.; Gong, Y.; Zou, Y.; He, Y.; Xiong, Y.; Luo, Y.; You, Y.; Liu, Y.; Zhou, Y.; Zhu, Y. X.; Xu, Y.; Huang, Y.; Li, Y.; Zheng, Y.; Zhu, Y.; Ma, Y.; Tang, Y.; Zha, Y.; Yan, Y.; Ren, Z. Z.; Ren, Z.; Sha, Z.; Fu, Z.; Xu, Z.; Xie, Z.; Zhang, Z.; Hao, Z.; Ma, Z.; Yan, Z.; Wu, Z.; Gu, Z.; Zhu, Z.; Liu, Z.; Li, Z.; Xie, Z.; Song, Z.; Pan, Z.; Huang, Z.; Xu, Z.; Zhang, Z.; Zhang, Z.; DeepSeek; AI. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. *arXiv* **2025**, arXiv:2501.12948.

(32) Lin, C.-Y. ROUGE: A Package for Automatic Evaluation of Summaries. *Text Summarization Branches Out*; Association for Computational Linguistics: Barcelona, Spain, 2004; pp 74–81.

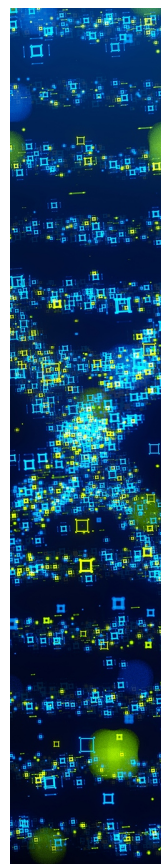
(33) Shaib, C.; Barrow, J.; Sun, J.; Siu, A. F.; Wallace, B. C.; Nenkova, A. Standardizing the Measurement of Text Diversity: A Tool and a Comparative Analysis of Scores. *arXiv* **2025**, arXiv:2403.00553.

(34) Quan, S. Automatically Generating Numerous Context-Driven SFT Data for LLMs across Diverse Granularity. *arXiv* **2024**, arXiv:2405.16579.

(35) Qwen; Yang, A.; Yang, B.; Zhang, B.; Hui, B.; Zheng, B.; Yu, B.; Li, C.; Liu, D.; Huang, F.; Wei, H.; Lin, H.; Yang, J.; Tu, J.; Zhang, J.; Yang, J.; Yang, J.; Zhou, J.; Lin, J.; Dang, K.; Lu, K.; Bao, K.; Yang, K.; Yu, L.; Li, M.; Xue, M.; Zhang, P.; Zhu, Q.; Men, R.; Lin, R.; Li, T.; Tang, T.; Xia, T.; Ren, X.; Ren, X.; Fan, Y.; Su, Y.; Zhang, Y.; Wan, Y.; Liu, Y.; Cui, Z.; Zhang, Z.; Qiu, Z. Qwen2.5 Technical Report. *arXiv* **2025**, arXiv:2412.15115.

(36) Liu, A.; Feng, B.; Xue, B.; Wang, B.; Wu, B.; Lu, C.; Zhao, C.; Deng, C.; Zhang, C.; Ruan, C.; Dai, D.; Guo, D.; Yang, D.; Chen, D.; Ji, D.; Li, E.; Lin, F.; Dai, F.; Luo, F.; Hao, G.; Chen, G.; Li, G.; Zhang, H.; Bao, H.; Xu, H.; Wang, H.; Zhang, H.; Ding, H.; Xin, H.; Gao, H.; Li, H.; Qu, H.; Cai, J. L.; Liang, J.; Guo, J.; Ni, J.; Li, J.; Wang, J.; Chen, J.; Chen, J.; Yuan, J.; Qiu, J.; Li, J.; Song, J.; Dong, K.; Hu, K.; Gao, K.; Guan, K.; Huang, K.; Yu, K.; Wang, L.; Zhang, L.; Xu, L.; Xia, L.; Zhao, L.; Wang, L.; Zhang, L.; Li, M.; Wang, M.; Zhang, M.; Zhang, M.; Tang, M.; Li, M.; Tian, N.; Huang, P.; Wang, P.; Zhang, P.; Wang, Q.; Zhu, Q.; Chen, Q.; Du, Q.; Chen, R. J.; Jin, R. L.; Ge, R.; Zhang, R.; Pan, R.; Wang, R.; Xu, R.; Zhang, R.; Chen,

- R.; Li, S. S.; Lu, S.; Zhou, S.; Chen, S.; Wu, S.; Ye, S.; Ye, S.; Ma, S.; Wang, S.; Zhou, S.; Yu, S.; Zhou, S.; Pan, S.; Wang, T.; Yun, T.; Pei, T.; Sun, T.; Xiao, W. L.; Zeng, W.; Zhao, W.; An, W.; Liu, W.; Liang, W.; Gao, W.; Yu, W.; Zhang, W.; Li, X. Q.; Jin, X.; Wang, X.; Bi, X.; Liu, X.; Wang, X.; Shen, X.; Chen, X.; Zhang, X.; Chen, X.; Nie, X.; Sun, X.; Wang, X.; Cheng, X.; Liu, X.; Xie, X.; Liu, X.; Yu, X.; Song, X.; Shan, X.; Zhou, X.; Yang, X.; Li, X.; Su, X.; Lin, X.; Li, Y. K.; Wang, Y. Q.; Wei, Y. X.; Zhu, Y. X.; Zhang, Y.; Xu, Y.; Xu, Y.; Huang, Y.; Li, Y.; Zhao, Y.; Sun, Y.; Li, Y.; Wang, Y.; Yu, Y.; Zheng, Y.; Zhang, Y.; Shi, Y.; Xiong, Y.; He, Y.; Tang, Y.; Piao, Y.; Wang, Y.; Tan, Y.; Ma, Y.; Liu, Y.; Guo, Y.; Wu, Y.; Ou, Y.; Zhu, Y.; Wang, Y.; Gong, Y.; Zou, Y.; He, Y.; Zha, Y.; Xiong, Y.; Ma, Y.; Yan, Y.; Luo, Y.; You, Y.; Liu, Y.; Zhou, Y.; Wu, Z. F.; Ren, Z. Z.; Ren, Z.; Sha, Z.; Fu, Z.; Xu, Z.; Huang, Z.; Zhang, Z.; Xie, Z.; Zhang, Z.; Hao, Z.; Gou, Z.; Ma, Z.; Yan, Z.; Shao, Z.; Xu, Z.; Wu, Z.; Zhang, Z.; Li, Z.; Gu, Z.; Zhu, Z.; Liu, Z.; Li, Z.; Xie, Z.; Song, Z.; Gao, Z.; Pan, Z.; DeepSeek-AL. DeepSeek-V3 Technical Report. *arXiv* **2025**, arXiv:2412.19437.
- (37) Zeng, A.; Xu, B.; Wang, B.; Zhang, C.; Yin, D.; Zhang, D.; Rojas, D.; Feng, G.; Zhao, H.; Lai, H.; Yu, H.; Wang, H.; Sun, J.; Zhang, J.; Cheng, J.; Gui, J.; Tang, J.; Zhang, J.; Sun, J.; Li, J.; Zhao, L.; Wu, L.; Zhong, L.; Liu, M.; Huang, M.; Zhang, P.; Zheng, Q.; Lu, R.; Duan, S.; Zhang, S.; Cao, S.; Yang, S.; Tam, W. L.; Zhao, W.; Liu, X.; Xia, X.; Zhang, X.; Gu, X.; Lv, X.; Liu, X.; Liu, X.; Yang, X.; Song, X.; Zhang, X.; An, Y.; Xu, Y.; Niu, Y.; Yang, Y.; Li, Y.; Bai, Y.; Dong, Y.; Qi, Z.; Wang, Z.; Yang, Z.; Du, Z.; Hou, Z.; Wang, Z.; Team GLM. ChatGLM: A Family of Large Language Models from GLM-130B to GLM-4 All Tools. *arXiv* **2024**, arXiv:2406.12793.
- (38) Tan, T. F.; Elangovan, K.; Jin, L.; Jie, Y.; Yong, L.; Lim, J.; Poh, S.; Ng, W. Y.; Lim, D.; Ke, Y.; Liu, N.; Ting, D. S. W. Fine-Tuning Large Language Model (LLM) Artificial Intelligence Chatbots in Ophthalmology and LLM-Based Evaluation Using GPT-4. *arXiv* **2024**, arXiv:2402.10083.
- (39) Hu, X.; Ru, D.; Qiu, L.; Guo, Q.; Zhang, T.; Xu, Y.; Luo, Y.; Liu, P.; Zhang, Y.; Zhang, Z. RefChecker: Reference-Based Fine-Grained Hallucination Checker and Benchmark for Large Language Models. *arXiv* **2024**, arXiv:2405.14486.
- (40) Si, S.; Zhao, H.; Chen, G.; Gao, C.; Bai, Y.; Wang, Z.; An, K.; Luo, K.; Qian, C.; Qi, F.; Chang, B.; Sun, M. Aligning Large Language Models to Follow Instructions and Hallucinate Less via Effective Data Filtering. *arXiv* **2025**, arXiv:2502.07340.
- (41) Mai, H. T.; Chu, C. X.; Paulheim, H. Do LLMs Really Adapt to Domains? An Ontology Learning Perspective. *arXiv* **2024**, arXiv:2407.19998.
- (42) Shukla, A. K. Domain-Specific LLMs, Retrieval-Augmented Generation, and Agentic AI: A Unified Architecture for Specialized Intelligence. *Social Science Research Network: Rochester, NY*, 2025.
- (43) Wang, C.; Liu, X.; Yue, Y.; Tang, X.; Zhang, T.; Jiayang, C.; Yao, Y.; Gao, W.; Hu, X.; Qi, Z.; Wang, Y.; Yang, L.; Wang, J.; Xie, X.; Zhang, Z.; Zhang, Y. Survey on Factuality in Large Language Models: Knowledge, Retrieval and Domain-Specificity. *arXiv* **2023**, arXiv:2310.07521.
- (44) Tang, L.; Laban, P.; Durrett, G. MiniCheck: Efficient Fact-Checking of LLMs on Grounding Documents. *arXiv* **2024**, arXiv:2404.10774.
- (45) Zhang, L.; Jijo, K.; Setty, S.; Chung, E.; Javid, F.; Vidra, N.; Clifford, T. Enhancing Large Language Model Performance To Answer Questions and Extract Information More Accurately. *arXiv* **2024**, arXiv:2402.01722.
- (46) Michael, M.; Neal, J.; Catastrophic, C. Interference in Connectionist Networks: The Sequential Learning Problem. *Psychology of Learning and Motivation*; Academic Press, 1989; Vol. 24; pp 109–165.
- (47) Yeo, E.; Tong, Y.; Niu, M.; Neubig, G.; Yue, X. Demystifying Long Chain-of-Thought Reasoning in LLMs. *arXiv* **2025**, arXiv:2502.03373.
- (48) He, H.; Rong, Z.; Ji, K.; Li, C.; Huang, Q.; Xia, C.; Yang, L.; Zhang, H. Rethinking Reasoning Quality in Large Language Models through Enhanced Chain-of-Thought via RL. *arXiv* **2025**, arXiv:2509.06024.
- (49) Yao, Z.; Liu, Y.; Chen, Y.; Chen, J.; Fang, J.; Hou, L.; Li, J.; Chua, T.-S. Are Reasoning Models More Prone to Hallucination? *arXiv* **2025**, arXiv:2505.23646.
- (50) Tran, K.-T.; Dao, D.; Nguyen, M.-D.; Pham, Q.-V.; O'Sullivan, B.; Nguyen, H. D. Multi-Agent Collaboration Mechanisms: A Survey of LLMs. *arXiv* **2025**, arXiv:2501.06322.
- (51) Qin, Y.; Hu, S.; Lin, Y.; Chen, W.; Ding, N.; Cui, G.; Zeng, Z.; Huang, Y.; Xiao, C.; Han, C.; Fung, Y. R.; Su, Y.; Wang, H.; Qian, C.; Tian, R.; Zhu, K.; Liang, S.; Shen, X.; Xu, B.; Zhang, Z.; Ye, Y.; Li, B.; Tang, Z.; Yi, J.; Zhu, Y.; Dai, Z.; Yan, L.; Cong, X.; Lu, Y.; Zhao, W.; Huang, Y.; Yan, J.; Han, X.; Sun, X.; Li, D.; Phang, J.; Yang, C.; Wu, T.; Ji, H.; Liu, Z.; Sun, M. Tool Learning with Foundation Models. *arXiv* **2024**, arXiv:2304.08354.
- (52) Schick, T.; Dwivedi-Yu, J.; Dessì, R.; Raileanu, R.; Lomeli, M.; Zettlemoyer, L.; Cancedda, N.; Scialom, T. Toolformer: Language Models Can Teach Themselves to Use Tools. *arXiv* **2023**, arXiv:2302.04761.
- (53) Xing, J.; Zheng, S.; Ding, D.; Kelly, J. T.; Wang, S.; Li, S.; Qin, T.; Ma, M.; Dong, Z.; Jang, C.; Zhu, Y.; Zheng, H.; Ren, L.; Liu, T.-Y.; Hao, J. Deep Learning for Prediction of the Air Quality Response to Emission Changes. *Environ. Sci. Technol.* **2020**, *54* (14), 8589–8600.
- (54) Qwen2.5 Speed Benchmark - Qwen. https://qwen.readthedocs.io/en/v2.5/benchmark/speed_benchmark.html (accessed Sept 25, 2025).
- (55) Tahmid, S.; Sarker, S. Qwen2.5-32B: Leveraging Self-Consistent Tool-Integrated Reasoning for Bengali Mathematical Olympiad Problem Solving. *arXiv* **2024**, arXiv:2411.05934.
- (56) Xu, H.; Wu, X.; Wang, W.; Li, Z.; Zheng, D.; Chen, B.; Hu, Y.; Kang, S.; Ji, J.; Zhang, Y.; Guo, Z.; Yang, Y.; Zhang, M.; Zhang, D. RedStar: Does Scaling Long-CoT Data Unlock Better Slow-Reasoning Systems? *arXiv* **2025**, arXiv:2501.11284.



CAS BIOFINDER DISCOVERY PLATFORM™

STOP DIGGING THROUGH DATA —START MAKING DISCOVERIES

CAS BioFinder helps you find the
right biological insights in seconds

Start your search



A division of the
American Chemical Society